MIL-R-36557C
20 November 1984
SUPERSEDING
MIL-R-36557B
2 February 1973

MILITARY SPECIFICATION

REGULATOR, PRESSURE, MEDICAL GAS ADMINISTRATION APPARATUS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the single-stage type regulator used to control pressure and flow rate, designed for use with oxygen type inhalators, as well as other oxygen therapy apparatus.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2) the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-P-116	- Preservation-Packaging, Methods of.
MIL-T-5542	- Thread Compound, Antiseize and Sealing, Oxygen Systems.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101) by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6680

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNDERWRITERS' LABORATORIES, INC. (UL)

UL 252 Gas Pressure Regulators
UL 404 High-Pressure Gas Gauges

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingsten Road, Northbrook, IL 60062; 1285 Walt Whitman Road, Melville, LI, NY 11746; and 1655 Scott Boulevard, Santa Clara, CA 95050.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B57.1 Compressed Gas Cylinder Valve Outlet and Inlet Connections.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

COMPRESSED GAS ASSOCIATION, INC. (CGA)

CGA Pamphlet V5 Diameter-Index Safety System (Medical Applications); Non-Interchangeable Low Pressure Connections For Medical Gases, Air and Suction.

(Application for copies should be addressed to the Compressed Gas Association, Inc., 1235 Jefferson Davis Highway, Arlington, VA 22202.)

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION (NASA)

NHB 8060.1B Flammability, Odor, and Offgassing Requirements and Test Procedures for Materials in Environments that Support Combustion.

(Application for copies should be addressed to the National Aeronautics and Space Administration, Washington, DC 20546.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 The contractor shall submit proof, prior to delivery, that the items to be delivered under this specification conform to the applicable requirements of UL Standards of Safety 252 and 404. Proof shall consist of one of the following:

a. A UL label affixed to the product, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the UL report directly to the contracting officer, or the government inspector, upon request.

b. A laboratory report from a nationally recognized independent laboratory If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The Government reserves the right to require a facilities report and to conduct a facilities survey. Ary laboratory report submitted shall include actual test methods, data taken, and results of tests. Detailed findings of examinations necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test sample. Compliance with these requirements does not excuse compliance with all other requirements of the contract and specifications.

3.2 Oxygen compatibility. All materials (metallic and nonmetallic) used in the construction of the regulator shall comply with requirements of NASA NHB 8060.1B for Group I, Type D material, except delete "NASA Center" or "NASA approval" and substitute "DPSC approval" wherever NASA approval is specified. The contractor shall provide a complete list of all materials used in the construction of the regulator to the contracting officer.

3.2.1 Material. The body of the regulator shall be fabricated from forged or machined copper alloy, nickel-plated or chromium-plated over nickel-plated; Monel Alloy 400 or Inconel 600. There shall be no internal plating of the regulator body. Alloys shall be tempered and shall have a minimum tensile strength of 50,000 pounds per square inch. All external brass or copper alloy surfaces, such as gland nut, standard threaded fitting for low-pressure oxygen connection and captive tubing adapter shall be nickel-plated or chromium-plated over nickel-plated. Material used in the construction of the

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regulator shall be of the highest quality throughout, suitable in all respects for the purpose intended. Alloys of aluminum, magnesium, or titanium shall not be used for metal components in direct communication with oxygen. Polytetrafluoroethylene (Teflon), polychlorotrifluoroethylene (KEL-F), or vinylidenefluoride hexafluoropropylene (Viton or Fluorel) shall be the only acceptable polymeric materials used in the internal construction of the regulators. All other plastic or rubber compounds shall be prohibited.

NOTE: It is recommended that the internal surfaces of the regulator body not be plated; if internal plating exists, flaking or other visible indications of poor plating adhesion shall be considered a defective regulator.

3.3 Style, design and dimensions. The regulator shall consist, essentially, of a body, a pressure-reducing valve, an inlet pressure gage, a single-pressure delivery device to indicate flow, a flow control valve for manually controlling the rate of low-pressure flow, a standard externally-threaded fitting for connection to low-pressure oxygen devices and a captive tubing adapter. The weight of the complete unit shall not exceed 2-1/2 pounds. The unit shall be suitable for use with oxygen cylinders and oxygen therapy equipment. Dimensions of the complete unit shall be in accordance with Figure 1.

3.3.1 Regulator. Regulator shall be of the single stage type, which shall maintain a uniform discharge rate by means of a sensitive diaphragm. The regulator shall have a flow capacity of not less than 15 liters per minute with 90 percent oxygen cylinder content depletion. A suitable safety device shall be incorporated, to prevent accumulation of hazardous pressures within the regulator, which shall discharge within a pressure range of 75 to 200 pounds per square inch gage (psig). The regulator shall be equipped with a commercial union nut and gland inlet connection which shall be capable of making connection with suitable oxygen cylinders having a 0.903-14 NGO right-hand thread on the outlet valve in strict accordance with Connection No. 540 of ANSI B57.1. The regulator inlet shall have a suitable, assembled filter with openings not greater than 66 microns or equivalent mesh, fabricated from bronze to prevent foreign matter from entering the device.

3.3.2 Low-pressure oxygen fitting and captive tubing adapter. A removable captive tubing adapter shall be attached to the low-pressure (outlet) side of the regulator. The tubing adapter shall be attached to a standard threaded oxygen fitting for low-pressure application, in accordance with CGA Connection No. 1240, as specified in CGA Pamphlet V5. This standard fitting shall make a tight, leakproof connection with the regulator body; the tubing adapter shall provide a tight, leakproof connection for rubber tubing with an inside diameter of 3/16 inch and a wall thickness of 1/16 to 3/32 inch. The configuration of the tubing adapter is shown for general style and design only.

3.3.3 Pressure gage. The regulator shall be equipped with a pressure gage having a minimum capacity of 3000 psig. The gage shall be calibrated to indicate both pressure and content ("1/4", "1/2", "3/4" and "FULL") remaining in an oxygen cylinder. The face cover of the gage shall be fabricated from a clear, durable, plastic.

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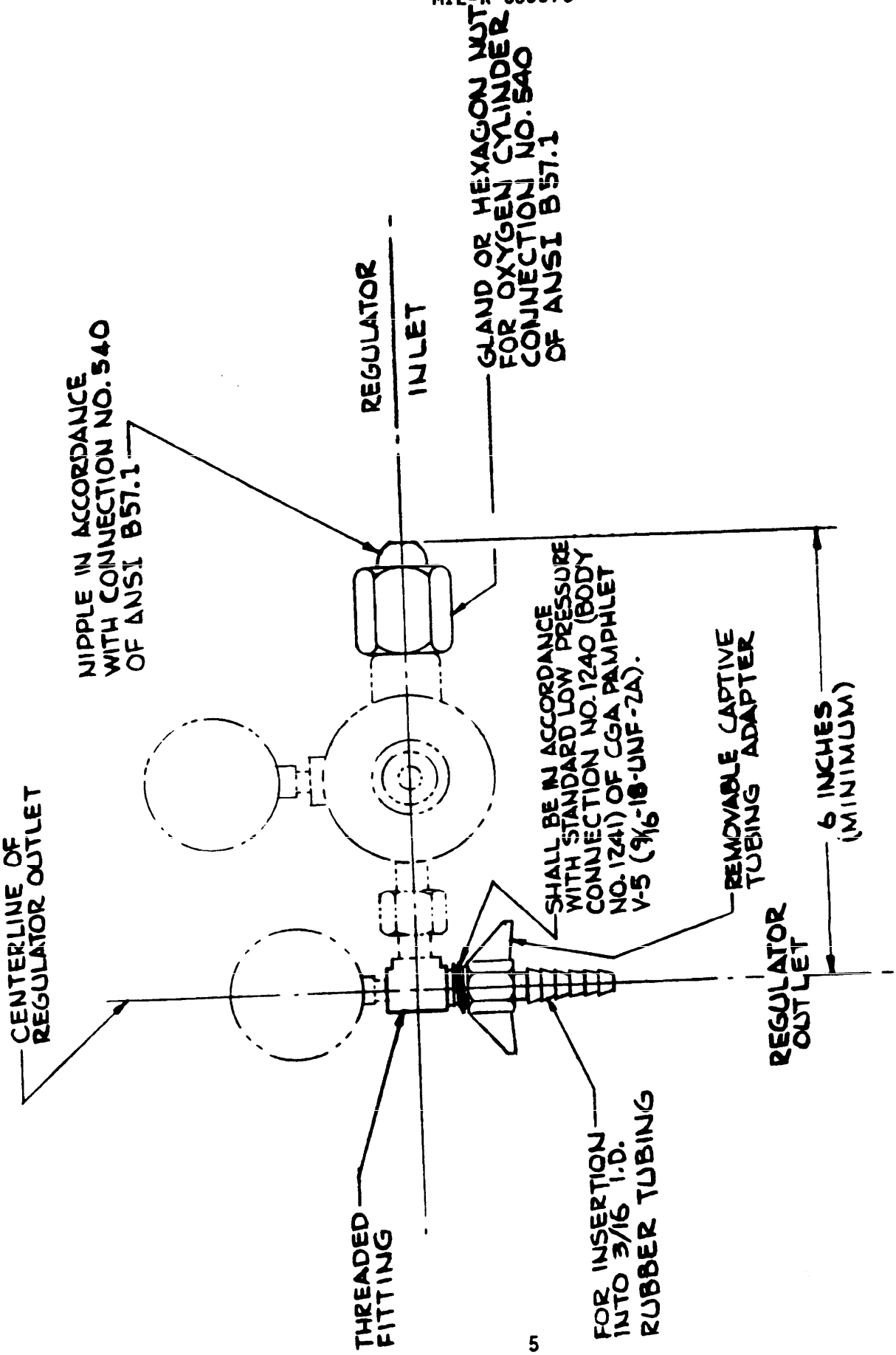


FIGURE 1.

Clearance required between regulator inlet and regulator outlet.

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3.3.4 Flow indicating device. The regulator shall be equipped with a Bourdon tube delivery pressure gage calibrated to indicate flow rates of no flow to 15 liters per minute of oxygen. The unit shall be capable of maintaining a constant flow rate at any flow rate setting from no flow to 15 liters per minute by means of a single control. The delivery pressure scale shall be graduated in liter per minute (LPM) increments starting at the 3 LPM graduation or lower LPM value. Numeral markings shall be indicated at each odd or even LPM graduation up to at least 12 LPM or the markings calibrated from the 2 LPM graduation up to and including, the 15 LPM graduation. The construction of the flow scale shall be such that the flow rate may be readily observed at a distance of 5 feet. The face cover of the delivery pressure flow scale shall be fabricated from a clear, durable, plastic.

3.3.5 Flow gage accuracy. When tested as specified in 4.4.1, the actual flow delivered by the regulator shall be within 1.5 LPM of the indicated flow of the regulator flowmeter during the discharge of 90 percent of the fully charged contents of a standard size oxygen cylinder. The allowable flow rate error may be bilateral, providing the total variance does not exceed 1.5 LPM at the settings specified in 4.4.1, for the total sample size. The delivery pressure gage shall return to the no flow position when all pressure is released or removed from the regulator. Minimum sample size for this test shall be 13 regulators for each production run or submitted lot.

3.3.6 Regulator leakage. When tested as specified in 4.4.2, the regulator shall not exhibit leakage as evidenced by the formation of bubbles.

3.4 Thread sealants. Organic-based thread sealants which are considered to be flammable when in contact with oxygen shall not be used. Thread sealants used for antiseize and sealing purposes shall be Teflon thread-sealing tape or thread compounds in strict accordance with MIL-T-5542. Thread sealing tape shall be applied so that it is not permitted to enter the oxygen flow stream. Thread sealing tape shall be applied with at least one internal male thread uncovered.

3.5 Service data. The contractor shall enclose, with each unit, two copies each of one or more booklets which contain complete instructions for installation, operation, maintenance, and a list of component parts. The service manual shall contain the following precautionary note:

"CAUTION: DO NOT USE ORGANIC-BASED THREAD SEALANTS ON ANY PORTION OF THE REGULATOR. USE ONLY TEFLON THREAD-SEALING TAPE OR 'THREAD COMPOUND, ANTISEIZE AND SEALING, OXYGEN SYSTEMS, 4 OZ. TUBE, NSN 8030-00-243-3284', AVAILABLE FROM THE GENERAL SERVICES ADMINISTRATION."

3.6 Identification marking. The regulator and/or the pressure gage and/or the flowmeter shall be clearly and legibly marked with the manufacturer's name or registered trademark.

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3.7 Workmanship. Regulators shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I. The unit of product for examination purposes shall be one complete regulator.

TABLE I. Sampling for examination.

Type defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics. The unit of product for test purposes shall be one complete regulator.

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TABLE II. Sampling for tests.

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL
Regulator	Dimensions	3.3	---	S-3	0.65
	Flow gage accuracy	3.3.5	4.4.1	S-3	1.0
	Leakage test	3.3.6	4.4.2	S-3	1.0

4.3 Examination. The regulators shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any regulator in the sample containing one or more defects shall be rejected and, if the number of defective regulators in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

TABLE III. Classification of defects.

Categories	Defects
Major	
101	External brass surfaces not plated.
102	Certification of Underwriters' Laboratories (or other Government-approved laboratory) missing.
103	External surfaces not free form corrosion, scale, pits, holes, and crevices.
104	Regulator gage calibration markings not as specified.
105	Flowmeter scale calibration markings not as specified.
106	Tubing adapter not furnished.
107	Regulator cannot be easily assembled to large (H-size) oxygen cylinder.
108	Complete unit weight exceeds 2-1/2 pounds.
109	Regulating device not easily adjusted.
110	Regulator bonnet not securely attached to regulator body.
111	One or more components loose (excluding tubing adapter).
112	Face cover of pressure gage and flowmeter not fabricated from clear plastic.
Major	
113	Regulator inlet does not possess filter.
114	Regulator inlet or flowmeter outlet blocked with foreign matter.
115	Service data not furnished or does not contain caution note.
Minor	
201	Manufacturer's identification markings missing.

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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Flow gage accuracy test. This test shall be performed in carefully controlled surroundings to insure maximum operator safety. A pneumatic circuit shall be assembled which shall consist, essentially, of the regulator to be tested (sample), a standard oxygen, compressed nitrogen, or compressed air cylinder with at least 90 percent of full charge remaining, an adjustable calibrated regulator and pressure gage with at least 3000 psig capacity, a calibrated precision flow-readout device to measure actual flow output of the sample, and necessary delivery tubing.

The adjustable, calibrated regulator shall be inserted between the pressure source and the sample. The outlet of the sample shall be connected to the calibrated flowmeter. All connections shall be made in a leakproof manner.

The test shall be conducted by regulating the pressure source to 2000, 1500, 1000, 500, and 250 psig with the sample regulator flow-adjusting screw or knob set at 5, 10, and 15 LPM respectively. Each test shall consist of recording the actual output flows of the sample in LPM by regulating the pressure source with the calibrated, adjustable regulator, in descending values; and subsequently recording the actual output flows of the sample by regulating the pressure source in ascending values. This procedure will result in obtaining two actual output flow values of the sample for each of the above regulated pressures and sample flow settings.

4.4.2 Leakage test. The leakage test shall be performed, essentially, by utilizing the same circuit specified in 4.4.1, except that the precision flowmeter may be removed from the circuit. The oxygen cylinder shall be capable of delivering at least 2000 psig pressure to the regulator. The regulator flow-adjusting screw or knob shall be turned out fully, and a film of soapy water (grease-free) shall be applied to all regulator junctions with 2000 psig internal pressure applied to the regulator. The regulator output shall then be suitably plugged. The regulator flow-adjusting screw or knob shall be turned all the way in (turn clockwise) and the regulator cap shall be tested for leaks with the use of grease-free, soapy water when 2000 psig internal pressure is applied to the regulator. As an alternate method, the entire regulator may be immersed in water at a depth no greater than 6 inches.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit. One regulator, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table IV shall be packaged in each unit, intermediate and exterior container, as applicable, for the required level of protection specified in the procurement document.

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TABLE IV. Packaging quantities.

Unit package	Intermediate package	Exterior container
1 unit	Not required	6 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Cleaning and drying. Each unit shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.3.2 Preservative. Preservatives shall not be applied.

5.1.3.3 Intimate wrap. Each unit shall be wrapped in neutral tissue or other suitable anticorrosive material, except that volatile corrosion inhibitors shall not be used.

5.1.3.4 Unit package. Each unit shall be packaged in accordance with MIL-P-116, method IC-2. As an alternate, each unit shall be wrapped in a suitable cushioning material, inserted into a heat-sealed plastic bag and packaged in a snug fitting box of an appropriate design.

5.1.4 Level C. Each unit shall be packaged in a standard commercial container of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-601, overseas type or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

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5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-601, domestic type or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking. Each unit package and shipping container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The regulator described herein shall be used in conjunction with oxygen type humidifiers and other oxygen therapy apparatus.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.4 This specification covers NSN 6680-00-935-4242 REGULATOR, PRESSURE, MEDICAL GAS ADMINISTRATION APPARATUS, Lightweight, Oxygen, with Flowmeter, appearing in the Federal Supply Catalog.

6.5 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:

DoD-MB

Agent:

DLA-DM

Project No. 6680-0186

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-36557C	REGULATOR, PRESSURE, MEDICAL GAS ADMINISTRATION APPARATUS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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